Spacing SF: Examining Inequity in Access to Essential Amenities

Group 9: Dynamic Duo!
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Track A

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Problem & Motivation

Urban environments distribute essential resources unevenly, often reflecting underlying socioeconomic inequalities. Our project asks: *How are key neighborhood amenities distributed across socioeconomically different areas of San Francisco, and do these spatial patterns produce compounded disadvantage?* We define amenities as essential services that support daily life and long-term opportunity, specifically food access (food services and SNAP retailers), public transit (Muni) and public schools. We also move beyond surface-level mapping and point counts, recognizing that not all amenities carry equal weight. To capture this complexity, we draw on accessibility literature to ensure a weighting schema that reflects real-world value, such as distinguishing between food vendors by type and transit stops by connectivity. Furthermore, we introduce three sub-questions to deepen our spatial understanding of amenity accessibility: (1) Food: Which tracts have lower-quality food access when food options are weighted by type based on service? (2) Transit: Which tracts have worse effective transit access when stops are weighted by route frequency and destination quality? (3) Schools: (3) Which tracts are more likely to score poorly across a combined food, transit, and school accessibility index simultaneously? The basis for these questions stems from existing research that indicates that deprivation is not an individual phenomenon but a cumulative one, as exemplified by cases such as "The Case of Montréal's Missing Food Deserts[1]," as well as research done in San Francisco related to food insecurity/access[2].

The metrics used in this study would be accessibility indicators (distance-based and network-based), amenity density, and indices measuring "amenity richness" or deprivation. This methodology takes into account research evidence showing that disparities in transit access contribute to the widening inequality gap[3][4] that access to cars creates significantly more opportunities than transit access, and that school location and funding mechanisms define long-term opportunities[5]. The unit of analysis chosen will be the census tract, which will enable an approximation of neighborhoods' differences while being compatible with socioeconomic data sources. Low-income people, urban planners, policymakers, and organizations focusing on urban equality are some of the main stakeholders. Spatial disparity in resource access may lead to poverty traps and reduce social mobility chances. This phenomenon has considerable consequences for society, whether from an analytical or practical standpoint.

Data

All original and modified versions of the data are available on our Github.

Census: The census tract dataset outlines subdivisions of San Francisco County based on how census data is collected, as of 2020. The income dataset contains the 2020 - 2024 5-year estimates of these individual census tracts. The census raw data had a few issues: the geometry was stored as plain text instead of a usable map format, the ID fields used to join the two datasets didn't match because of extra characters in the ACS/income dataset, and income values were stored as text rather than

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[1] Apparicio P, Cloutier MS, Shearmur R. The case of Montréal's missing food deserts: evaluation of accessibility to food supermarkets. Int J Health Geogr. 2007 Feb 12;6:4. doi: 10.1186/1476-072X-6-4. PMID: 17295912; PMCID: PMC1803774.

[2] Ramirez, Yesenia, "Food Insecurity in San Francisco (Examination of some local and state interventions)" (2022). *Master's Projects and Capstones*. 1485.

[3] Akbar, Prottoy A. 2024. "Public Transit Access and Income Segregation." CEPR Discussion Paper 19558.

[4] Maharjan, S., Janatabadi, F., & Ermagun, A. (2024). Spatial Inequity of Transit and Automobile Access Gap across America for Underserved Population. Transportation Research Record: Journal of the Transportation Research Board, 2678(1), 674-690.

[5] Zheng, Angela, and James Graham. 2022. "Public Education Inequality and Intergenerational Mobility." American Economic Journal: Macroeconomics 14 (3): 250–82.

numbers. There was also an empty column with no values. To fix these, the geometry was converted and reprojected to a coordinate system suited for San Francisco (ESPG: 2227), the IDs were standardized so the datasets could be joined correctly, columns were cleaned up and renamed, and a couple of offshore island tracts (the Farallons) were removed since they skewed the map. No duplicate records were found, the joined dataset came out complete with no missing values, and the final cleaned file was saved as a GeoJSON for use in the rest of the project.

Data + Description	**Coverage**	**Granularity**	**Type**	**Format**	**Source**
Census Tract	San Francisco County, 2020	Citywide-Level, Multipolygon Coordinates, Individual Tracts	Continuous	CSV	DataSF
Income	San Francisco County, 2020 - 2024	Median Value of Individual Census Tracts	Discrete	CSV	US Census Bureau

Food: The Active Food Services dataset documents various types of eateries (restaurants, bars, cafés, etc.) in different neighborhoods in San Francisco. Some of these services we will not consider in our analysis due to its restriction to openness to the public (i.e. employee-only cafeterias), and will continue to be cleaned. The SNAP retailers dataset indicates places that accept SNAP[6] as a form of payment; lastly, the Mobile Food Services dataset lists food trucks and push carts, both with active and expired permits. The common issue with the raw data in these datasets was that they included locations that had expired permits or no longer existed. Exact duplicates were removed, and a majority of the data cleaning involved removing the locations that were not active in the relevant time frame. For the SNAP Retailers and Mobile Food Services datasets, there were no latitude and longitude columns, so in the future we will attempt to chart the locations based on its given address.

Data + Description	**Coverage**	**Granularity**	**Type**	**Format**	**Source**
Active Food Services	San Francisco, 1960 - Present	Citywide-Level, Individual Businesses	Categorical	CSV	DataSF
SNAP Retailers	San Francisco, 2020	Citywide-Level, Individual Businesses, Addresses	Categorical	CSV	PolicyMap
Mobile Food Services	San Francisco, 2015 - Present	Citywide-Level, Individual Businesses, Addresses	Categorical	CSV	Open Data SF

Transit: **Important to note: as a duo team, we assigned one person to work on 1 amenity type to clean and analyze for the midterm. Our transit data is still in progress.** The transit dataset, which focuses on Muni, details the system's stopping locations and frequency. For the future documentation process, we will exclude stops that are no longer in service, cross check any inconsistencies with the given data, and only eliminate exact duplicate stops that are *not* for detour use.

Data + Description	**Coverage**	**Granularity**	**Type**	**Format**	**Source**
Muni Stops	San Francisco County, 2025	Citywide-Level, Latitude & Longitude, Individual Stops	Mixed[7]	CSV	DataSF

[6] Supplemental Nutrition Assistance Program.
[7] Contains categorical, discrete, and continuous data.

Schools: Public school location data for the 2024–25 academic year was sourced from the Institute of Education Sciences, which provides the name, address, and coordinates of all 132 public schools in San Francisco. The raw schools data was a national dataset, so the first step was filtering it down to San Francisco County only, which left 132 schools. Columns were then renamed for clarity and unnecessary fields were dropped. The cleaned dataframe was converted to a GeoDataFrame using the latitude and longitude columns, reprojected to match the SF tracts coordinate system (EPSG:2227), and confirmed to have no missing values before being saved as a GeoJSON. Finally, we calculated schools per census tract and distances from tract centroids to the nearest school.

Data + Description	Coverage	Granularity	Type	Format	Source
School Locations	San Francisco County, 2024-5	Citywide-Level, Latitude & Longitude	Continuous	CSV	NCES

Missing Data

Below, we summarize the missing information from each dataset (M), its connection to *Dark Data* (D), and how we are managing its absence (A).

Census + Income

- **M:** After merging the census boundary layer with the median income dataset, we found that 93% of median income information for census tracts was missing.
- **D:** When merging both, tracts with small populations were likely removed to protect privacy and treated as NaN (Type 1: Data We Know Are Missing. In addition, certain populations such as undocumented or unhoused residents are likely undercounted or missing (Type 4: Self Selection/Non-response).
- **A:** At this time, we left them as NaNs and colored grey so as to not make assumptions about suppressed census data.

Food

- **M:** The Active Food services dataset contains missing values about the location's end date, address information, and business corridor. The SNAP Retailers dataset interestingly contains no missing values. Lastly, the Mobile Food Services dataset has some locations with their facility type, expiration date, days and hours of operation, and the type of food sold missing.
- **D:** Data That We Know Is Missing. A majority of the other datasets have corresponding values that are not missing, and the information for those that *are* can be found with research.
- **A:** The missing values for these datasets are not influential, because we are only interested in locations that are still open and have a future expiration date. However, if the latitude and longitude for a location was missing, we would use the available address to geocode and find those values.

Transit

- **M:** None.
- **D:** Data That Changes With Time. Muni networks have expanded greatly since its inception in 1912, which *implies* greater access to various neighborhoods.
- **A**: If there were NA values, we would cross-check with another reputable source about the Muni Stops and impute the information from there.

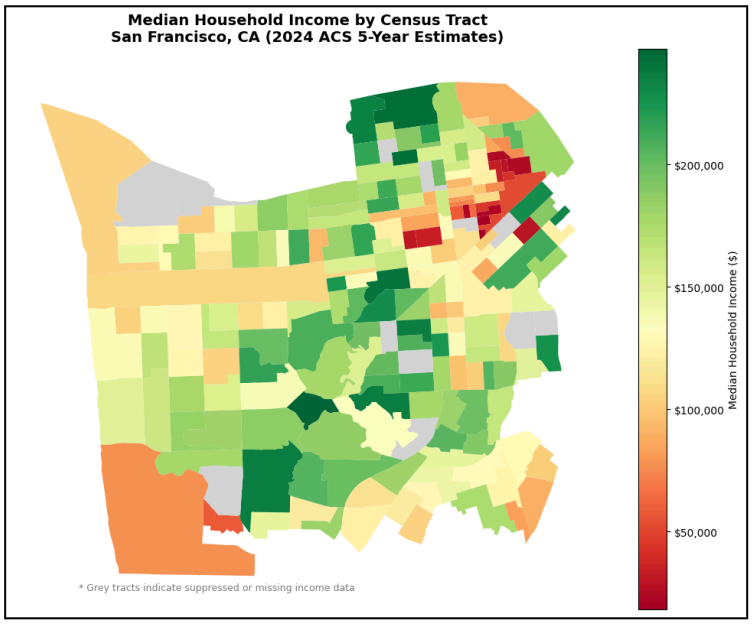
Schools

- **M:** None.
- **D:** While our school location dataset was cleaned without any missing data, the data only consists of public schools, leaving out private and charter schools (Type 3: Data Specifically Chosen). In addition, the data solely contains location information, limiting how much can be inferred about other measures of access (Type 5 Missing Important Unobserved Aspects).1
- **A:** N/A. If there were missing data, they would have been dropped.

Initial Results

Median Income by Census Tracts: **Map 1** shows median household income across San Francisco's census tracts, with greener areas indicating higher incomes and redder areas indicating lower incomes. San Francisco's median household income averages around $145,000 (SD = $54, 367) across census tracts. But when we look at the full range, the median income is as low as $18,217 in the poorest tract to $247,500 in the wealthiest. The map reflects this clearly, with wealthier neighborhoods concentrated in the north and lower-income areas spread across the center and southeast.

Map 1



***Food*:** Two of the three food datasets showed an interesting pattern. For the mobile food services (**Figure 1**), the small quantity of push carts are somewhat clustered in a certain latitude-longitude range, whereas food trucks are more scattered, although in a more centralized longitude range. For the eateries (**Figure 2)**, our data is currently highly skewed by establishments that have varying openness to the public (i.e. bars that serve food, but entry is only for those 21+). After cleaning further, we expect to observe notable patterns. Overall, the distribution of the type of food service may reflect the surrounding area's socioeconomic level, which we will investigate further in our next steps. After the initial cleaning, there are 160 remaining mobile food services (~25.6% of the original dataset) and 10,009 eateries that we are working with, ~23% of them being solely restaurants.

Figure 1

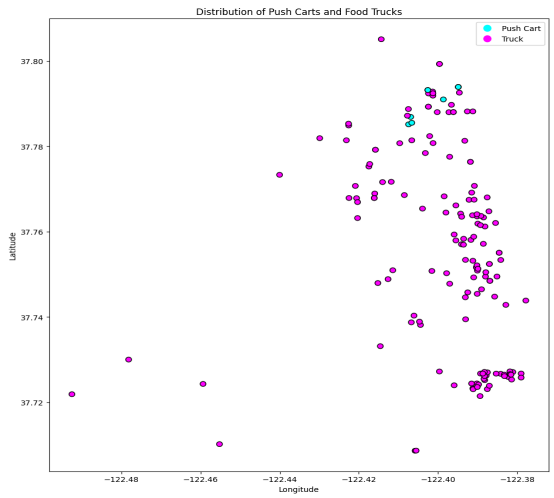
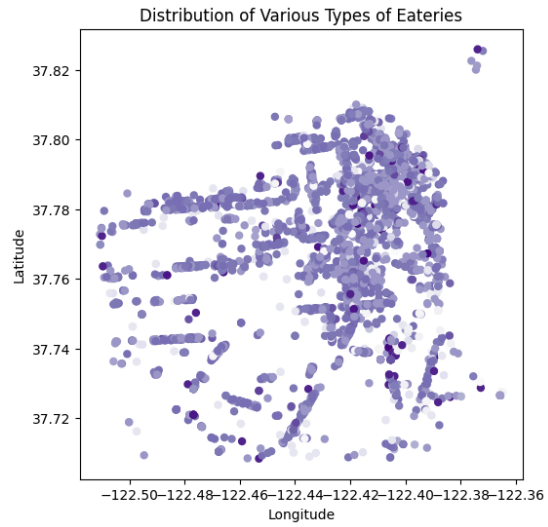
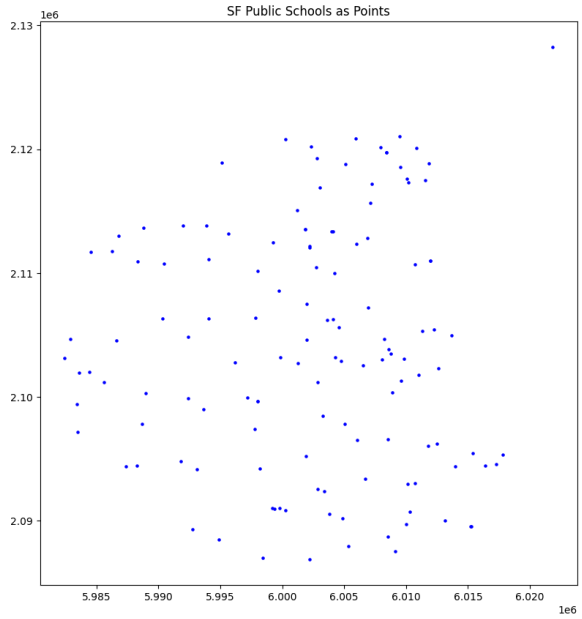


Figure 2

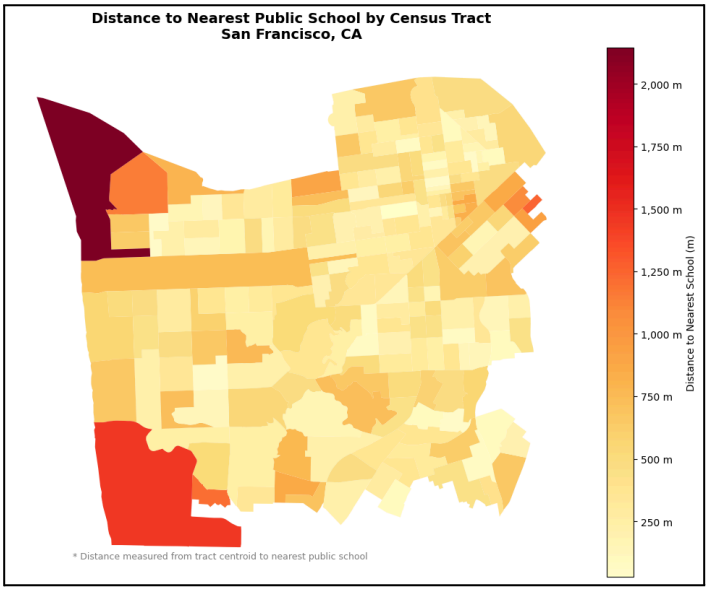


Schools: On average, census tracts are about 405 meters from the nearest public school, and half of all tracts are within 366 meters which is roughly a 5-minute walk, suggesting that public school access is generally pretty good across most of San Francisco. However, the map tells a more nuanced story in which the northwest corner has tracts over 2,000 meters from the nearest school, shown in dark red, while the school point map confirms that schools are much more densely clustered in the central and eastern parts of the city. The wide range from 28 meters to 2,146 meters suggests that while most residents are well served, a small number of tracts particularly in the west and southwest are significantly farther from public schools than the rest of the city.

Figure 3



Map 2



Progress & Plan to Completion

Since our initial proposal, we have taken the initiative to specify our main research question, *What is the distribution of public amenities among socioeconomically diverse census tracts in San Francisco?* Specifically, we developed three subquestions that have an outcome that can be measured quantitatively and investigate if the result is associated with the socioeconomic status of the respective census tract. Additionally, the amenities that we originally subjectively identified as important are now supported with historical research literature. These subquestions are:

1. **Food:** Are low-income tracts more likely to have only low-quality food options (like limited assortment markets) while higher-income tracts have access to full-service supermarkets?
2. **Transit**: Do Muni routes in low-income tracts actually connect residents to grocery stores and schools, or just to other low-income areas?
3. **Schools**: Are low-income tracts more likely to sit in "amenity deserts" where school access is poor AND food and transit access are poor simultaneously?

For our remaining journey to completion, our tasks include creating all interactive maps, visually polishing them and ensuring that they are accessible, organizing our work in the Github repository, collaborating on the creation of transit analysis, and creating the ArcGIS website. As a team of two, we made the helpful decision to use our midterm token so that we can have additional time to put in significant effort. In combination with RRR week, this helps our timeline to submission be feasible and realistic. After receiving midterm feedback, we anticipate spending 3 weeks on the aforementioned tasks and submitting our final work at least 24 hours before the deadline to ensure that there are no last-minute technical issues. Below is our tentative schedule until submission:

- **Week of April 13th:** While waiting for feedback, reconvene about our next tasks.
- **Week of April 20th:** Integrate feedback into our creations.
- **Week of April 27th:** Continue working on tasks and getting additional feedback in office hours.
- **Week of May 4th:** Troubleshoot any issues and submit by May 7th!

References

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